

PHASE OFF!

The three digital advantages—robust transmission, identical-to-the-original copies and vastly improved editing—put the tape-based analog technology at a severe disadvantage.

Pre DV: ANALOG	Post DV: DIGITAL
An Analog signal is vulnerable to interference that can significantly degrade picture quality. Timing errors can make straight vertical lines wavy in playback.	Digital signals are more forgiving when it comes to fluctuations in signal strength, and can incorporate checking of algorithms to ensure that a signal has been received as it was sent.
Quality degrades during copying. There is a significant loss of information associated with each transfer from one tape to another, so quality worsens with each generation.	Digital video suffers virtually no degradation in quality, no matter how many copies are made.
Every scene in a video project is shot several times. If you're a director trying to choose between take one and take six, you can't simply switch back and forth between them. You have to roll through takes two, three, four and five every time. Even with high-speed tape decks, this adds up to an enormous waste of time	In contrast, non-linear DV editing allows instant location and playback of any take of any scene. Non-linear editing has another advantage. Once you've picked the best take from each scene, you can assemble a—rough cut—in a few minutes just by deleting the scenes you rejected.

Video editing on a PC

This Typically requires a slightly beefed up home computer with a capture card, running software such as Avid's Media Composer, Apple's Final Cut Pro and iMovie, Adobe Premiere, or perhaps Broadcast 2000 if you're working on Linux.

Simple video editing software, like Studio, lets you organise clips in the same manner as photos in a photo album and assemble them in any order, by dragging and dropping. When you play the whole movie, the computer plays each clip, one after another.

More professional software, like Adobe Premiere, lets you arrange the clips on a more complex looking timeline and apply special effects like bluescreen transparency—an effect where a person appears against a fake background such as a solar system or a still photograph. With this you can create transitions, such as dissolving from one clip to another. Adding soundtracks is also possible.

The bare minimum ingredients for a competent DV-capable setup includes: 700 MHz computer, 128 MB of RAM, a 30 GB hard drive, a FireWire card, a video editing program, and a FireWire equipped DV camcorder.

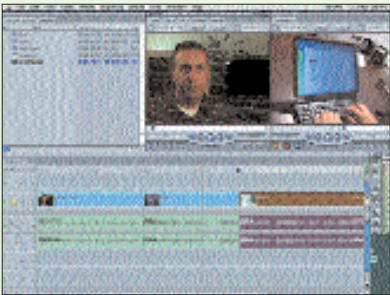
The images, by being digitised into the machine, can now be cut-pasted, warped, rendered, 'Photoshopped', and twisted around until they tell a tangible

tale. Digital video editing is used for television shows, TV commercials, music videos, and promotional movies.

Film effects can be achieved, to a certain degree, with Adobe Premiere. Another program, called Cinelook, can reproduce a wide array of film-looks—8, 16, and 35 mm film-stocks. It can give movies that gritty 'film-damage' look as well.

The Killer Apps

Final Cut Pro is a suite of editing programs rolled into one super-shiny package that just reeks of awesomeness with cutting, juggling, and every other type of special effect to keep a movie junkie happy. While Final Cut Pro doesn't have all the power of a standalone, professional sound editing program, it is quite adequate. Also, there's an additional track for any music you



Online movies

Over the last year, independent filmmakers have been putting their work online for the first time, with Internet start-ups such as Atom Films and Ifilm.com giving would-be filmmakers a place to showcase their newest works. Ifilm uses a democratic review process where the viewers determine which films move up the charts based on the number of times the movies are viewed.

DV and the Internet have created a new life for short films, which have never found viable outlets for mainstream audiences. Likewise, new distribution methods are allowing niche filmmakers to reach the audiences that might be interested in their peculiar tastes. Internet as a medium for movies has become an attractive alternative to the traditional distribution route.

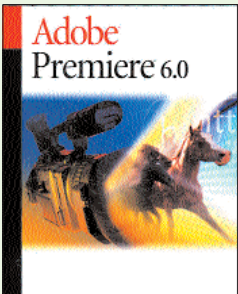
The future is virtual

All the pieces are in place to make DV products a huge market success, particularly when it comes to image capture and editing. To begin with, all the supporting technologies already exist. DV camcorders outsell conventional analog units in Japan.

The DV standard is backwards compatible—content creators can begin working with DV products now, without having to worry about the market penetration of a new platform for their work.

The DV market will almost certainly follow the pattern of numerous other technology-based products where the ini-

might want to import off CDs or any other source.



Adobe Premiere 6.0 can handle all the cut or transition edits, apply rolling titles and credits, and add soundtracks. The resulting production can then be export-

ed to tape or a variety of computer files such as AVI, MPEG, QuickTime, and RealMedia. Premiere 6.0 can also work directly over FireWire with a DV machine for importing and exporting video. Very handy indeed!

